

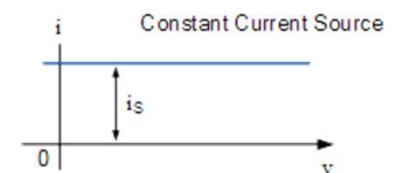
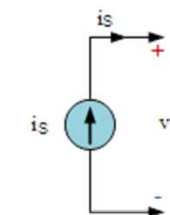
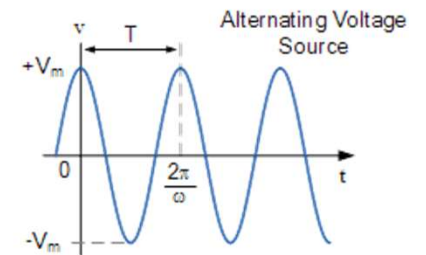
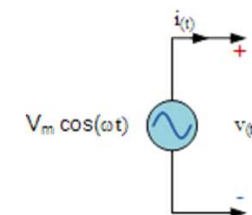
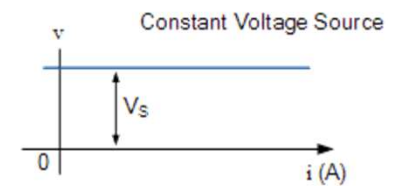
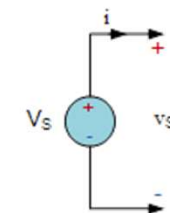
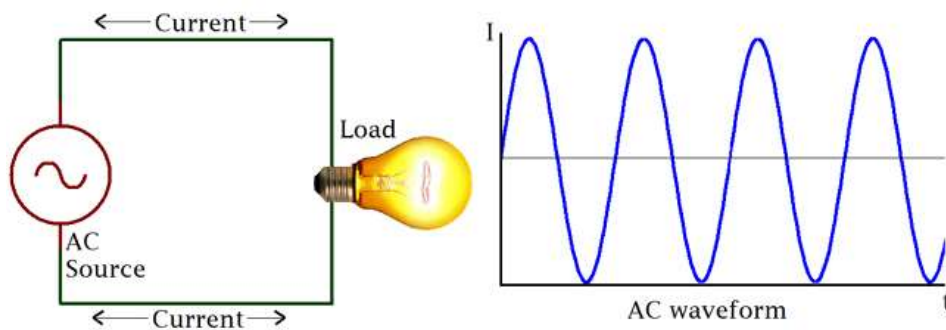
AC Circuits

Waveforms and RMS Value

Power and Power factor

What is Alternating Voltages and Currents

A voltage which changes its polarity at regular intervals of time is called alternating voltage.



Sinusoidal Alternating Voltages and Currents

Commercial alternator produce sinusoidal alternating voltage.

The sinusoidal alternating voltage can be expressed by the equation:

$$a = A_{\max} \sin(\omega t)$$

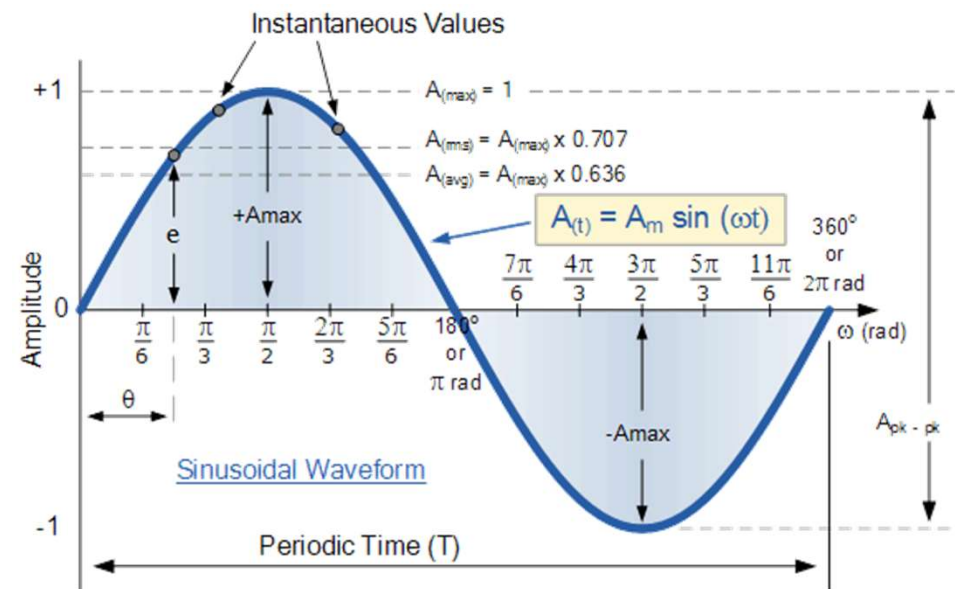
$$a = A_{\max} \sin(2\pi ft)$$

ω = Angular velocity

a = Instantaneous value of alternating voltage

$A_{\max}$ = Maximum value of alternating voltage

f = Frequency





Why Sine Waveform ?

- Sine waveform produces the least disturbance in the electrical circuit and is the smoothest and efficient waveform.
- The mathematical computations, connected with alternating current work, are much simpler with this wave.

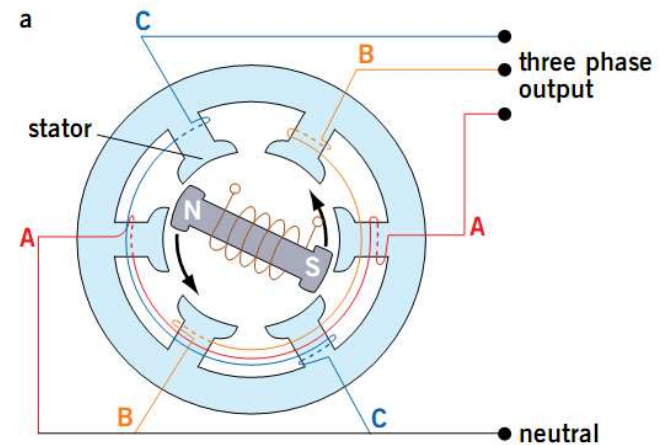
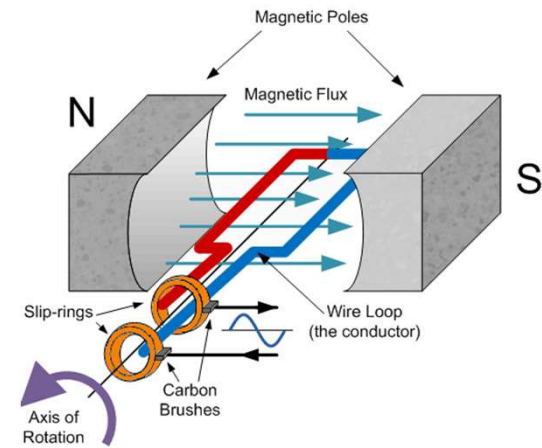
Generation of Alternating Voltages and Currents

An alternating voltage may be generated:

i) By rotating a coil at constant angular velocity in a uniform magnetic field.

OR

ii) By rotating a magnetic field at a constant angular velocity within a stationary coil.





Equation of alternating voltages and currents

Let the time be measured from the instant of coincidence of the plane of the coil with the X-axis. At this instant maximum flux, $\Phi_{\max}$ links with the coil. Let the coil assume the position with ωt , as shown in Fig. , after moving in anticlockwise direction for t seconds.

The angle θ through which the coil has rotated in t seconds = ωt

In this position,

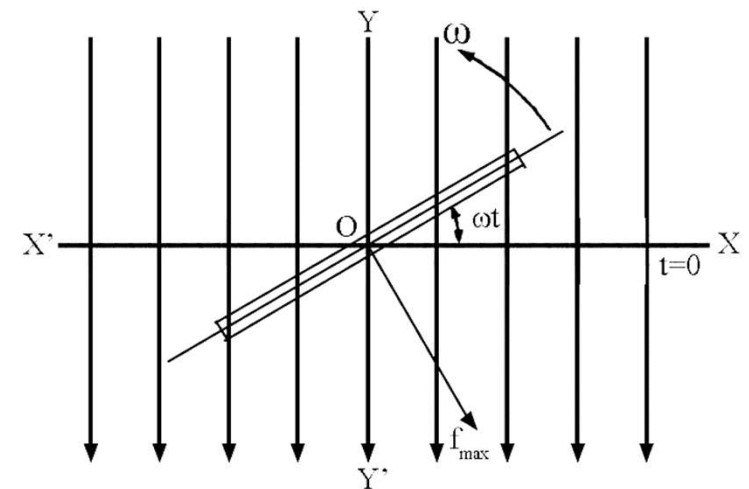
the component of flux along perpendicular to the plane of coil = $\Phi_{\max} \cos \omega t$.

Hence flux linkages of the coil at this instant = Number of turns on coil $\times$ linking flux

i.e; instantaneous flux linkages = $N \Phi_{\max} \cos \omega t$.

Since emf induced in a coil is equal to the rate of change of flux linkages with minus sign,

EMF induced at any instant,



Rotation of Rectangular
Coil in Uniform Magnetic Field



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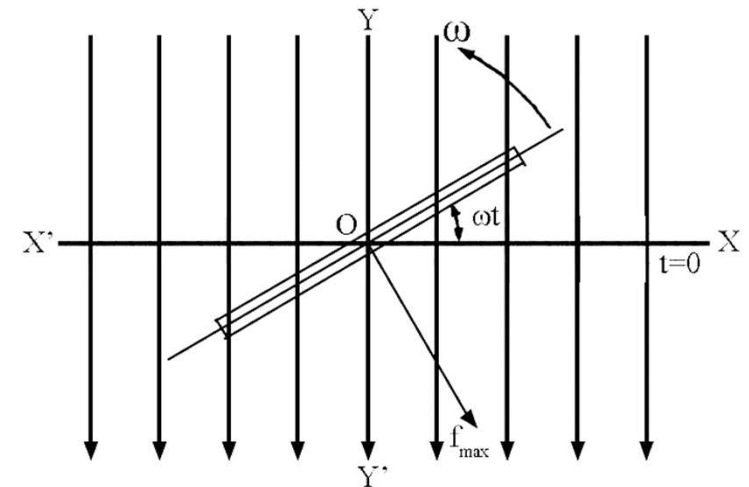
EMF induced at any instant,

$$\begin{aligned} e &= -\frac{d}{dt}[N\phi_{\max}\cos\omega t] \\ &= \phi_{\max} N \frac{d}{dt}[-\cos\omega t] = \phi_{\max} N \omega \sin\omega t \\ e &= E_{\max} \sin\omega t \end{aligned}$$

$$e = NBA\omega \sin\omega t$$

When $\omega t = 0$, $\sin \omega t = 0$, therefore, induced emf is maximum, which is denoted by $E_{\max}$ and is equal to $\Phi_{\max} N \omega$

Substituting $\Phi_{\max} N \omega = E_{\max} = NBA\omega$



Rotation of Rectangular
Coil in Uniform Magnetic Field



Lenz's law is contained in the rigorous treatment of Faraday's law of induction (the magnitude of EMF induced in a coil is proportional to the rate of change of the magnetic flux), where it finds expression by the negative sign:

$$\mathcal{E} = - \frac{d\Phi_{\mathbf{B}}}{dt},$$

which indicates that the induced electromotive force and the rate of change in magnetic flux have opposite signs.



Problem:

An AC generator consists of a coil of 100 turns and cross sectional area of 3m^2 , rotating at a constant angular speed of 60 rad per second in a uniform magnetic field of 0.04 T. The resistance of the coil is 500 ohm. Calculate maximum voltage, current and power. Ans: 1036.8W



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Solution

$$N = 100, A = 3\text{m}^2, \omega = 60\text{rad/s}$$

$$B = \quad R =$$

As induced emf is given by $= NBA\omega \sin\omega t$

$$\text{Max emf } e_0 = NBA\omega =$$

$$= \quad \text{Volt}$$

$$\text{Induced current (max) } I_0 = \frac{e_0}{R} = \quad = 18\text{A}$$

$$\text{Power} = e_0 I_0 = \quad \text{watt}$$

Important AC Terminology

Waveform:

Instantaneous value:

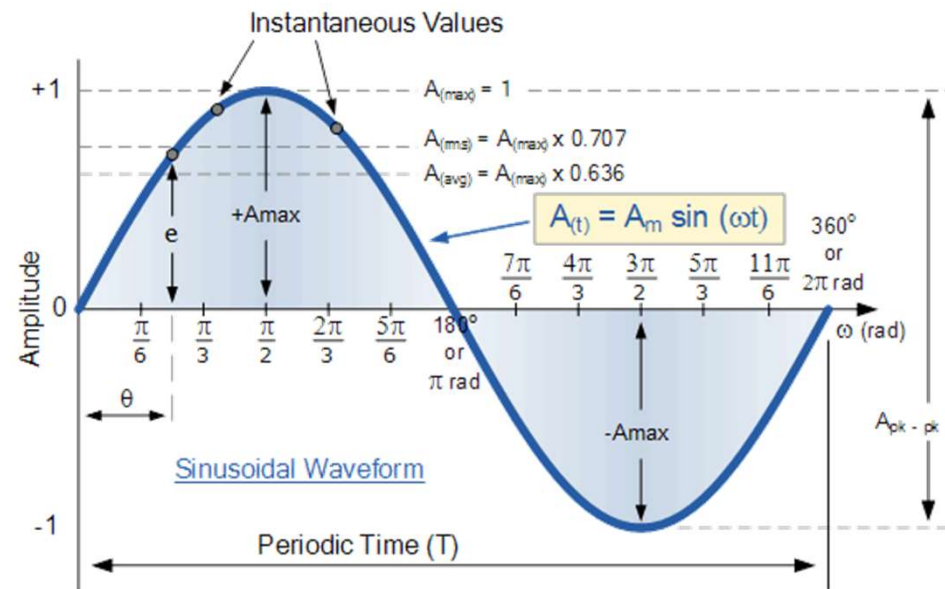
Cycle:

Time period:

Frequency:

Angular velocity:

Amplitude:





Important AC Terminology

Frequency and speed:

Consider a coil rotating at a speed of N r.p.m. in the field of P poles. As the coil moves past successive north poles, one complete cycle is generated. In one revolution of the coil, $P/2$ cycle will be generated.

Frequency

$f = \text{Number of cycles per second}$

$$f = \frac{\text{Number of cycles}}{\text{Revolution}} \times \frac{\text{Revolutions}}{\text{Second}}$$

$$f = p \times n$$

$$f = \frac{P}{2} \times \frac{N}{60}$$

$$f = \frac{PN}{120}$$

$$N = 120f/P$$

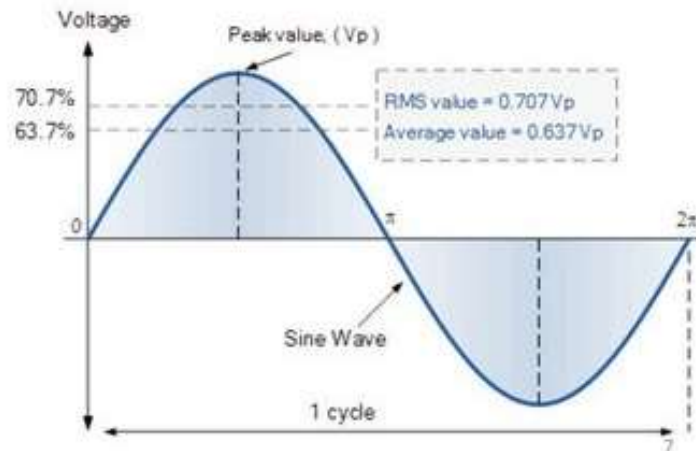
Average value of Sinusoidal Voltages and Currents

- In a pure sine wave if the average value is calculated over the full cycle, the average value would be equal to zero as the positive and negative halves will cancel each other out.
- Then the average value is obtained by adding the instantaneous values of voltage over one half cycle only.

$$V_{av} = \frac{1}{\pi} \int_0^{\pi} V_p \sin \theta \, d\theta$$

$$V_{av} = \frac{V_p}{\pi} [-\cos \theta]_0^{\pi}$$

$$V_{av} = \frac{2V_p}{\pi} = 0.637V_p$$





Problem

Express each of the following frequencies in Hertz

- a) 40 cycles in 4.0 seconds
- b) 80 cycles in 200 milliseconds
- c) 1000 revolutions in 0.5 seconds
- d) 600 rotations in 1 minute



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Solution:

- a) $40/4.0 = 10$ cycles per second = 10 Hz
- b) $80/0.2 = 400$ cycles per second = 400 Hz
- c) $1000/0.5 = 2000$ cycles per second = 2000 Hz (2 kHz)
- d) $600/60 = 10$ cycles per second = 10 Hz



Problem

Express each of the following as angular velocity in radians per second

- a) 80 rad in 10 s
- b) 2.5 krad in 50 s
- c) 400 rad in 200 s
- d) 40 Mrad in 10 s



Problem

Express each of the following as angular velocity in radians per second

- a) 80 rad in 10 s
- b) 2.5 krad in 50 s
- c) 400 rad in 200 s
- d) 40 Mrad in 10 s

Solution:

- a) $\omega = 80/10 = 8 \text{ rad/s}$
- b) $\omega = 2500/50 = 500 \text{ rad/s}$
- c) $\omega = 400/200 = 2.0 \text{ rad/s}$
- d) $\omega = (40 \times 10^6)/10 = 4.0 \text{ rad/s}$



Problem

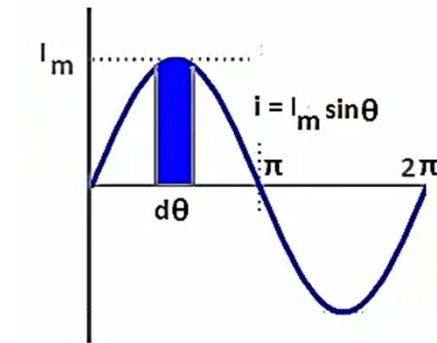
An alternating current is expressed as $I = 14.14 \sin 314t$. Determine:- i) Maximum current, ii) Frequency, iii) Time period.

ANS: i) 14.14 A ii) 50 Hz iii) 0.02 S

RMS or Root-mean-square or Effective value

The term *rms* stands for (square) root of the mean of the squares of the instantaneous current values.

$$\begin{aligned}
 \therefore I_{rms} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} i^2 \cdot d\theta} \\
 I_{rms}^2 &= \frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta \, d\theta = \frac{I_m^2}{\pi} \int_0^{\pi} \sin^2 \theta \cdot d\theta \\
 &= \frac{I_m^2}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) \cdot d\theta = \frac{I_m^2}{2\pi} \int_0^{\pi} (1 - \cos 2\theta) \cdot d\theta \\
 &= \frac{I_m^2}{2\pi} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi} \\
 &= \frac{I_m^2}{2\pi} \left[\left(\pi - \frac{1}{2} \sin 2\pi \right) - \left(0 - \frac{1}{2} \sin 2 \times 0 \right) \right] \\
 &= \frac{I_m^2}{2\pi} \left[\pi - 0 - 0 + 0 \right] = \frac{\pi}{2\pi} I_m^2 = \frac{I_m^2}{2} \\
 \therefore I_{rms} &= \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m
 \end{aligned}$$





Few points on RMS values

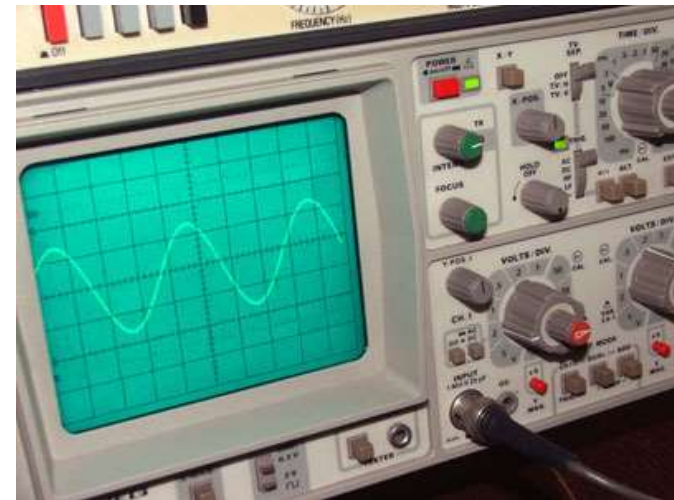
$$v = V_{\max} \sin(\omega t)$$

$$V_{p-p} = 2 \times V_{\max}$$

$$V_{\text{avg}} = V_{\max} / \pi = 0.637 \times V_{\max}$$

$$V_{\text{RMS}} = V_{\max} / \sqrt{2} = 0.707 \times V_{\max}$$

$$V_{\max} = \sqrt{2} \times V_{\text{RMS}} = 1.414 \times V_{\text{RMS}}$$





Problems

An alternating current is expressed as $v=20 \sin(2 \times 314 t)$, Determine the i) Maximum voltage, ii) RMS voltage, iii) Time period, iv) Frequency.

A sinusoidal current has maximum value of 650 mA. What is its average value and RMS value?

The equation of alternating voltage is given by $v = 325.22 \sin 314t$. Find (i) RMS value (ii) Frequency (iii) Average value.



Form factor and Peak Factor

Form factor: The ratio of r.m.s. value to the average value of an alternating quantity is known as form factor.

$$\begin{aligned}\text{Form Factor} &= \frac{V_{rms}}{V_{av}} \text{ or } \frac{I_{rms}}{I_{av}} \\ &= \frac{V_{rms}}{V_{av}} \\ FF &= \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}}\end{aligned}$$

Where V_m is the peak value of AC voltage

$$\text{Form Factor} = \frac{\pi V_m}{2\sqrt{2}V_m}$$

$$\text{Form Factor} = 1.11$$



Form factor and Peak Factor

Peak factor: The ratio of maximum value to the r.m.s. value of an alternating quantity is known as peak factor.

– for **any waveform** the peak factor is defined as

$$\text{Peak factor} = \frac{\text{peak value}}{\text{r.m.s. value}}$$

– for a **sine wave** this gives

$$\text{Peak factor} = \frac{V_p}{0.707 V_p} = 1.414$$



Problems:

- The maximum current in a sinusoidal a.c. circuit is 10A. What is the instantaneous current at 60° ?

ANS: 8.66 A

- Write the mathematical expression for a 50 Hz sinusoidal voltage of peak value 80 V. Sketch the waveform versus time t.
- An alternating current of frequency 60 HZ has a maximum value 120 A.
 - i) Write down the equation for the instantaneous value.
 - ii) Find the instantaneous value of current after $1/360$ second. Consider initial current 0 A.
 - iii) Time taken to reach 96 A for the first time.



Power and Power factor:

Let $v = V_m \sin \omega t$ and $i = I_m \sin (\omega t \pm \phi)$

Therefore, the instantaneous power is given by

$$\begin{aligned} p &= vi = V_m I_m \sin \omega t \sin (\omega t \pm \phi) \\ &= \frac{1}{2} V_m I_m [2 \sin \omega t \sin (\omega t \pm \phi)] \\ &= \frac{V_m I_m}{2} [\cos (\omega t - \omega t \mp \phi) - \cos (\omega t + \omega t \pm \phi)] \\ &= \frac{V_m I_m}{2} [\cos \phi - \cos (2\omega t \pm \phi)] \quad [\because \cos (-\phi) = \cos \phi] \end{aligned}$$

Average power over a cycle is given by

$$\begin{aligned} P_{av} &= \frac{V_m I_m}{2} \left[\frac{1}{T} \int_0^T \cos \phi \, dt - \frac{1}{T} \int_0^T \cos (2\omega t \mp \phi) \, dt \right] \\ &= \frac{V_m I_m}{2T} \cdot T \cos \phi = \frac{V_m I_m}{2} \cos \phi \\ &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi \quad \text{or} \quad P_{av} = V_{rms} I_{rms} \cos \phi \end{aligned}$$



Power and Power factor:

$$P_{av} = V_{rms} I_{rms} \cos \phi$$

The term $\cos \phi$ is known as the **power factor** of the given circuit. Thus,

$$\cos \phi = \frac{R}{Z} = \text{power factor} = \frac{\text{true power}}{\text{apparent power}}$$

For a pure resistive circuit, V and I are in phase ($\phi = 0^\circ$) hence, $\cos \phi = 1$ and average power $= V_{rms} I_{rms}$

For a pure inductive or a pure capacitive circuit, current and voltage differ in phase by $\frac{\pi}{2}$ (i.e. $\phi = \frac{\pi}{2}$).



Problems:

- An alternating current is given by –
 $i = 10 \sin 942 t$.
Determine the time taken from $t=0$ for the current to reach a value of 6 A for a first and second time.
- A sinusoidal voltage of 50 Hz has a maximum value of $200\sqrt{2}$ volts. At what time measured from a positive maximum value will the instantaneous voltage be 141.4 volts.



Problems:

- An alternating current, when passed through a resistor immersed in water for 5 minutes, just raised the temperature of water to boiling point. When a direct current of 4 A was passed through the same resistor under identical conditions, it took 8 minutes to boil the water. Find the r.m.s. value of the alternating current.



Problems:

- A current has the following steady values in amperes for equal intervals of time changing instantaneously from one value to the next. 0, 10, 20, 30, 20, 10, 0, -10, -20, -30, -20, -10, 0 etc. Calculate -i) average value ii) r.m.s value iii) form factor iv) peak factor.



Thank you